

Cycles of the Juggler map

Philippe Cochin
philippe@cochin.fr

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. We do not prove that conjecture.

A realized parity word w obeys the power envelope $(J^{|w|}(n))^{2^{|w|}} \leq n^{3^{\#O(w)}}$, so a cycle word is formally expanding. Inverse cells and two next-square thresholds give an elementary census: there is no nontrivial cycle of length at most seven. The leftover even-terminating two-even words form two infinite families, $O^{k-2}EE$ and $O^{k-3}EOE$, both excluded at every length $k \geq 6$. On a cycle minimum, a three-even leftover with a sufficiently long second gap reduces to those families. Seven bunched three-even families are likewise excluded. These are family exclusions, not a census at length eight or nine.

The same envelope, with the floor remainders kept, is an exact global defect. Zero defect characterizes the monochrome power towers. The defect is not a uniform per-step tax — none exists — but around a cycle it must finance the formal surplus $3^o - 2^L$. The resulting inequality, $n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o$ at a cycle minimum, restricts every remaining period to a near-convergent of $\ln 2 / \ln 3$, or to a huge length. Combined with a verified descent floor through 10^6 , there is no period at most 1053, and no period at most 10^5 outside an explicit set of 397 near-convergent lengths. The first record survivor is length 1054.

Even starts and odd-to-even starts have a uniform one- or two-step descent. The complementary class is the odd-to-odd starts. No density result is stated or used here.

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1. Introduction

The Juggler sequence was introduced by Pickover [1,2]. The one-step map is OEIS A094683 [3]; the stopping-time table is A007320 [4]. The orbit of 3 is 3, 5, 11, 36, 6, 2, 1. The orbit of 37 already peaks at 24906114455136. Universal arrival at 1 remains the open conjecture of the subject.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \quad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities.

The elementary first half of the note is a small-cycle census: there is no nontrivial cycle of period at most six (Theorem 3.6), and none of period at most seven (Theorem 3.8). The tool is the finite-word envelope (Theorem 2.2), together with the inverse cells and two next-square thresholds. After the census, the same cells exclude two infinite leftover families at every expanding length (Theorem 3.12), transport that comparison

across a first even letter on a cycle minimum (Theorem 3.13), exclude the seven bunched three-even families (Theorems 3.14–3.20), and upgrade both gapped leftovers from cycle minima to cycle words (Theorem 3.21). The word census of Theorems 3.6 and 3.8 stops at length seven; Section 5 excludes later periods by financing, not by a length-eight or length-nine word census.

The exact defect (Theorems 2.4–2.6) is the same recurrence with remainders kept rather than dropped. It classifies the rigid zero cases and shows that a uniform per-step slack tax is impossible. Around a cycle the surplus $\Delta_w(n) = n^{3^o} - n^{2^L}$ is exact (Corollary 2.7), and the remainders must finance it. The resulting inequality (Theorem 5.1) is

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o$$

at a cycle minimum. Combined with a verified descent floor through 10^6 , every period outside a near-convergent set of $\ln 2 / \ln 3$ is excluded through length 10^5 (Theorem 5.3). The first record survivor is length 1054.

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$. Write J^k for the k -fold iterate. A nonempty realized word w with $J^{|w|}(n) = n$ is a *cycle word*. The unique fixed point is 1; a cycle is *nontrivial* when it contains some $n \geq 2$. A cycle word at n is *minimum-based* when n is a cycle minimum: $J^j(n) \geq n$ for every $0 \leq j < |w|$. Every cycle word has a minimum-based rotation.

1.1 Related work

Pickover’s later exposition is Chapter 45 of [2]. Weisstein [5] records the map, the stopping-time sequences A094670, A094679, A095908, and a verification of arrival at 1 through 10^6 . That verification is used in Theorem 5.3, together with an independent first-passage computation recorded in Appendix B. OEIS A094716 [6] records extreme heights, including the start 48443 whose peak has 972 463 digits. Those height records do not bound the period.

We know of no published exclusion of nontrivial cycles for the exact floor-power map J . Prasad–Prasad [7] estimate excursion and stopping constants for juggler-like maps by a random-walk large-deviation model; those estimates do not apply here. Small-cycle censuses are a standard first layer for Collatz-like maps, surveyed by Lagarias [8,9]. Those results do not transfer: the branches of J are floor powers rather than affine maps (Crandall [10], Matthews–Watts [11]). In particular there is no identity of the form $n(2^K - 3^p) = C$.

The financing-versus-gap template of Section 5 follows Simons and de Weger [13] on Collatz m -cycles. The proof below is independent of that paper: floor-power defects are relatively $O(1/x)$ in logarithms, so one residual floor excludes every length that is not a near-convergent of $\ln 2 / \ln 3$. The leftover shape — period at least X , or one of a named convergent family — is the packaging used by Eliahou [14] for Collatz cycles.

Itinerary-class densities are the subject of a companion manuscript [12] and are not used here.

1.2 Verification

The arguments of Sections 2, 3, 5, and 6 may be read without machine assistance. An independent Lean check of those arguments lives in the repository named in Section 6; the corresponding theorem names are collected in Appendix A. Lemma 3.5 uses a table of 254 six-step evaluations for $2 \leq n < 256$, one table for each of the two leftover words. Lemma 3.7 uses a table of 12 seven-step evaluations for $2 \leq n < 14$. Lemma 3.11 uses the same window to record that no start $2 \leq n < 256$ realizes seven consecutive odd letters. Theorem 3.12 adds two 254-start tables at length eight and one at length nine; Theorem 3.14 reuses the length-nine table for *OOOOOEEE* below 128; Theorem 3.15 uses tables below 314 and below 16; Theorem 3.16 reuses the 254-start window at prefix lengths 4, 5, 6; Theorem 3.18 uses the same two windows as Theorem 3.15; Theorems 3.17, 3.19, and 3.20 reuse the 254-start window at the short expanding prefixes of those families. Those tables are finite computations, not a termination proof.

Theorem 5.3 uses an exact-integer first-passage run through 10^6 , together with an exact gap table of lengths up to 10^5 . Both live in the repository directory named in Appendix B. Weisstein’s published verification through 10^6 [5] is an independent check of the same floor. A companion Lean development, using a residual

floor of 261 and a census through length 19, concludes that the period is 84 or at least 85; that leftover is recorded in Appendix A and is not a theorem of the text. Length 84 is already excluded by Theorem 5.3.

2. Envelope and defect

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w . Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters. When w is realized, this endpoint is the $|w|$ -fold iterate.

Theorem 2.1 (fixed-word monotonicity). If $n \leq m$ and both realize w , then $J^{|w|}(n) \leq J^{|w|}(m)$.

Proof. Induct on w . The empty word is immediate. The images after a common realized prefix remain ordered, and both current states realize the same next letter. The even branch is the monotone integer square root; the odd branch is $x \mapsto x^3$ followed by the integer square root. $\square$

The realizing set of a fixed word need not be an interval: OE is realized at 7 and at 11 but not at 9.

Theorem 2.2 (finite-word power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Proof. The empty word is the equality $n \leq n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \leq n^{3^o}$, and the next letter is realized.

If the next letter is even, then $J(x)^2 \leq x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{2^\ell} \leq n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x)^2 \leq x^3$, so

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{3 \cdot 2^\ell} = (x^{2^\ell})^3 \leq (n^{3^o})^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. $\square$

Corollary 2.3 (exponent-gap contraction). If $n \geq 2$, w is realized at n , and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \leq n^{3^{\#O(w)}}$. The exponent gap and $n \geq 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \geq 1$, one has $m < n$. $\square$

The corollary includes familiar contracting blocks such as $OOOEE$. It does not prove that every start realizes some contracting word.

The floor slack is exact. For a single branch,

$$x^e = J(x)^2 + \rho(x), \quad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \leq \rho(x) < 2J(x) + 1$. Write

$$\text{gap}(a, \rho, e) = (a + \rho)^e - a^e,$$

so $a^e + \text{gap}(a, \rho, e) = (a + \rho)^e$. If $e \geq 1$, then $\text{gap}(a, \rho, e) = 0$ if and only if $\rho = 0$, and $\rho^e \leq \text{gap}(a, \rho, e)$.

Along a realized word w , write $x_0 = n$ and $x_{j+1} = J(x_j)$, and let $\rho_j = \rho(x_j)$. Define a running slack by $D_0 = 0$ and

$$D_{j+1} = \begin{cases} D_j + \text{gap}((x_{j+1})^2, \rho_j, 2^j), & \text{next letter even,} \\ \text{gap}((x_{j+1})^2, \rho_j, 2^j) + \text{gap}(x_j^{2^j}, D_j, 3), & \text{next letter odd.} \end{cases}$$

The *global defect* is $\Delta_w(n) = D_{|w|}$. An even letter keeps the old slack and lifts the new remainder through 2^j . An odd letter first cubes the running slack and then lifts the new remainder. In particular Δ is not the sum of the local remainders.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \quad \Delta_w(n) \geq 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \geq 0$.

Proof. The empty word is $n = n + 0$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $n^{3^o} = x^{2^\ell} + D$.

If the next letter is even, then $x = J(x)^2 + \rho(x)$, so

$$n^{3^o} = (J(x)^2 + \rho(x))^{2^\ell} + D = J(x)^{2^{\ell+1}} + D + \text{gap}(J(x)^2, \rho(x), 2^\ell).$$

The odd count is unchanged, and the new slack is the even update.

If the next letter is odd, then $x^3 = J(x)^2 + \rho(x)$. Cubing the inductive identity gives

$$n^{3^{o+1}} = (x^{2^\ell} + D)^3 = x^{3 \cdot 2^\ell} + \text{gap}(x^{2^\ell}, D, 3).$$

The leading term is

$$x^{3 \cdot 2^\ell} = (J(x)^2 + \rho(x))^{2^\ell} = J(x)^{2^{\ell+1}} + \text{gap}(J(x)^2, \rho(x), 2^\ell),$$

which is the odd update. $\square$

For a one-letter illustration, take $n = 3$ and $w = O$. Then $J(3) = 5$ and $3^3 = 27 = 5^2 + 2$, so $\Delta_O(3) = 2$. After a later letter the lift is a power-gap, not a sum of remainders.

Theorem 2.5 (vanishing). If w is realized at n , the following are equivalent.

1. $\Delta_w(n) = 0$.
2. Every local remainder along w vanishes.
3. $(J^{|w|}(n))^{2^{|w|}} = n^{3^{\#O(w)}}$.

In that case w is monochrome. More precisely, either $w = E^k$ and $n = a^{2^k}$ for an even a , or $w = O^k$ and $n = a^{2^k}$ for an odd a . A realized mixed word therefore has $\Delta_w(n) > 0$.

Proof. The identity of Theorem 2.4 gives (1) $\Leftrightarrow$ (3). For (1) $\Leftrightarrow$ (2), use the recurrence. If $e \geq 1$, a power-gap vanishes if and only if its addend vanishes. The even update is a sum of two nonnegative terms, and the odd update is a sum of two power-gaps; either vanishes only when the incoming slack and the new remainder both vanish. Thus $D_{|w|} = 0$ forces $D_0 = 0$ and every $\rho_j = 0$. The converse is immediate.

If every remainder vanishes, an even step satisfies $x_j = x_{j+1}^2$, while an odd step satisfies $x_j^3 = x_{j+1}^2$. In either case x_j and x_{j+1} have the same parity, so the itinerary is monochrome. If $w = E^k$, repeated substitution gives $n = x_k^{2^k}$; writing $a = x_k$, its parity is even. If $w = O^k$, unique factorization applied to $x_j^3 = x_{j+1}^2$ shows that every prime valuation of x_j is even. Iterating this valuation relation shows that every prime valuation of $n = x_0$ is divisible by 2^k , hence $n = a^{2^k}$ for an odd a . Conversely, these even and odd towers make every local remainder zero. $\square$

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then

$$\Delta_{uv}(n) = \text{gap}(m^{2^{|u|}}, \Delta_u(n), 3^{\#O(v)}) + \text{gap}((J^{|v|}(m))^{2^{|v|}}, \Delta_v(m), 2^{|u|}).$$

In particular $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$. Every local remainder satisfies $\rho_j^{2^j} \leq \Delta_w(n)$.

Proof. Write $o_u = \#O(u)$ and $o_v = \#O(v)$. Theorem 2.4 on u and on uv gives

$$n^{3^{o_u+o_v}} = (m^{2^{|u|}} + \Delta_u(n))^{3^{o_v}} = (J^{|uv|}(n))^{2^{|uv|}} + \Delta_{uv}(n).$$

The first power expands as

$$m^{2^{|u|} \cdot 3^{o_v}} + \text{gap}(m^{2^{|u|}}, \Delta_u(n), 3^{o_v}).$$

Theorem 2.4 on v at m rewrites the leading term as

$$\left((J^{|v|}(m))^{2^{|v|}} + \Delta_v(m) \right)^{2^{|u|}},$$

which expands as the second gap plus $(J^{|uv|}(n))^{2^{|uv|}}$. Comparing the two expressions for $n^{3^{o_u+o_v}}$ yields the identity. The suffix inequality is $\rho^e \leq \text{gap}(a, \rho, e)$ on the second summand. For a local remainder at index j , split w after j letters: the suffix defect is at least ρ_j , and raising through 2^j gives the stated bound. $\square$

Composition is polynomial, not additive: later odd letters raise the prefix slack to the third power. The naive recurrence $\Delta \leftarrow \Delta + \rho$ is false.

Corollary 2.7 (cycle surplus). If w is a cycle word at n , then

$$\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$$

exactly.

Proof. Theorem 2.4 with $m = n$. $\square$

The identity does not supply a state-independent positive tax, and none exists. For one realized letter at x with branch exponent $e \in \{1, 3\}$,

$$x^e < (J(x) + 1)^2,$$

so the relative slack $1 + \eta = x^e / J(x)^2$ satisfies

$$\eta < \frac{2}{J(x)} + \frac{1}{J(x)^2},$$

which tends to 0 as the state grows. The recorded extreme is an *OOE* block at $n = 180370579261640036336071806107777 \approx 1.80 \cdot 10^{32}$ whose word-relative slack

$$q_w(n) = \frac{n^{3^{\#O(w)}}}{(J^{|w|}(n))^{2^{|w|}}} - 1$$

satisfies $0 < q_{OOE}(n) < 10^{-30}$, by the exact integer comparison $0 < (n^9 - J^3(n)^8) 10^{30} < J^3(n)^8$.

3. Inverse cells and the census

The exact one-step fibers are

$$J(n) = q \iff q^2 \leq n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \leq n^3 < (m+1)^2 \quad (n \text{ odd}).$$

Lemma 3.1 (odd cells are unique). An odd fiber contains at most one integer. An even fiber is a parity-restricted square interval and may contain many predecessors.

Proof. Suppose $a < b$ lie in the same odd cell indexed by m . Then $(a+1)^3 \leq b^3 < (m+1)^2$ and $m^2 \leq a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

hence $3a^2 + 3a < 2m$ and $3a^2 < 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \leq a^3$ gives $9a^4 < 4m^2 \leq 4a^3$, contradicting $4a^3 < 9a^4$. $\square$

Theorem 3.2 (cycle restrictions). Let w be a cycle word at $n \geq 2$.

(i) The word is formally expanding:

$$2^{|w|} < 3^{\#O(w)}.$$

A contracting word cannot close a nontrivial cycle.

(ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter.

(iii) A realized word v is *superquadratic* if

$$3^{\#O(v)} \geq 2^{|v|+1}.$$

Any realized path from a start $n \geq 2$ to a state at least n^2 is superquadratic. On a cycle minimum the path to any later even state is superquadratic. The prefix OOE is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \leq n^{3^{\#O(w)}}$. Since $n \geq 2$, comparison of exponents gives $2^{|w|} \leq 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \geq 1$. Alternatively: a mixed cycle word has $\Delta_w(n) > 0$ by Theorem 2.5, so the envelope is strict; a monochrome tower cannot return for $n \geq 2$.

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start $n \geq 2$. An even state $x \geq 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \geq 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n . If the last letter were odd, the odd return cell would give $n^2 \leq x^3 < (n+1)^2$. Minimality gives $x \geq n$, hence $n^3 < (n+1)^2$, impossible for $n \geq 3$; and the minimum is odd, so $n \geq 3$.

For (iii), let a realized word v send n to $y \geq n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since $n \geq 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n . Thus $n \leq J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \leq y$, and the preceding argument applies to that prefix. $\square$

Lemma 3.3 (coarse lower envelope). For $n \geq 1$, write $q = \lfloor \sqrt{n} \rfloor$. Then $q^2 \leq n < (q+1)^2$. For $q \geq 1$ one has $(q+1)^2 \leq 4q^2$, and the second inequality is strict for $q \geq 2$, while for $q = 1$ one has $n < 4$. In all cases

$$n < 4 \lfloor \sqrt{n} \rfloor^2.$$

Thus an even step satisfies $n \leq 4J(n)^2$ and an odd step satisfies $n^3 \leq 4J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o} \leq C_v J^k(n)^{2^k}.$$

The constant starts at 1 and updates by $C \mapsto C \cdot 4^{2^j}$ on an even letter and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j . In particular $C_{OOO} = 2^{38}$ and $C_{OOOO} = 2^{130}$.

Proof. The comparison $n < 4q^2$ is the paragraph above. The composition law is the same recurrence as the proof of Theorem 2.2, with a factor 4^{2^j} inserted at each letter. The values C_{OOO} and C_{OOOO} are the result of that recurrence on those two words. $\square$

Lemma 3.4 (next-square thresholds). (i) If $q \geq 5$ realizes OO , then $J^2(q) \geq (q+1)^2$. (ii) If $q \geq 3$ realizes OOO , then $J^3(q) \geq (q+1)^2$. (iii) If a realized word v satisfies $J^{|v|}(q) \geq (q+1)^2$ and the next realized letter is odd, then $J^{|v|+1}(q) \geq (q+1)^2$. (iv) If vE is a cycle word at n , then $J^{|v|}(n) < (n+1)^2$. (v) If $a \geq 3$, then $O^a E$ is not a cycle word at any $n \geq 2$.

Proof. For (i), write $m = \lfloor q^{3/2} \rfloor$. The bound $J^2(q) \geq (q+1)^2$ follows from $m^3 \geq (q+1)^4$, because then $\lfloor m^{3/2} \rfloor \geq (q+1)^2$. If $q = 5$, then $11^2 = 121 \leq 125 = 5^3 < 144 = 12^2$, so $m = 11$, and $11^3 = 1331 \geq 6^4 = 1296$. Since q realizes OO , it is odd, so the only remaining case is $q \geq 7$. Then $m \geq q^{3/2} - 1$, so it is enough that $(q^{3/2} - 1)^3 \geq (q+1)^4$. Expanding the left side and dropping the positive remainder $3q^{3/2} - 1$ reduces this to

$q^{9/2} - 3q^3 \geq (q+1)^4$, or equivalently $\sqrt{q} - 3/q \geq (1 + 1/q)^4$. The left side increases for $q > 0$ and the right side decreases, so the case $q = 7$ suffices:

$$\sqrt{7} - \frac{3}{7} > \frac{5}{2} - \frac{3}{7} = \frac{29}{14} > \frac{4096}{2401} = \left(\frac{8}{7}\right)^4,$$

where $\sqrt{7} > 5/2$ follows from $25/4 < 7$.

For (ii), the orbit $3 \rightarrow 5 \rightarrow 11 \rightarrow 36$ gives $J^3(3) = 36 \geq 16$. If $q \geq 5$ realizes OOO , then it realizes OO , so (i) gives $J^2(q) \geq (q+1)^2$. The third letter is odd, hence $J^3(q) = J(J^2(q)) \geq J^2(q)$.

For (iii), the image after v is odd and at least $(q+1)^2 \geq 4$, so the odd branch does not decrease it.

For (iv), the last letter is even, so the preimage $z = J^{|v|}(n)$ is even and satisfies $n^2 \leq z < (n+1)^2$.

For (v), suppose $O^a E$ is a cycle word at $n \geq 2$. The start is odd, hence at least 3, and realizes O^a . Parts (ii) and (iii) give $J^a(n) \geq (n+1)^2$, contradicting (iv). $\square$

Lemma 3.5 (two length-six exclusions). Neither $OOOEEOE$ nor $OOOOEE$ is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2 \cdot 256^{64}$ because $(1+1/256)^{64} < e^{64/256} = e^{1/4} < 2$, and $256(n+1) \leq 257n$, so $(n+1)^{64} < 2n^{64}$. Then $2^{130}(n+1)^{64} < 2^{131}n^{64}$. Since $n \geq 256 = 2^8$, one has $n^{17} \geq 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \leq n^{81}$.

For $2 \leq n < 256$, neither word returns to its start. This is a table of 254 evaluations of each word: at every start, either some letter fails to match the current parity, or the six-step image differs from the start. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooee` and `no_cycle_word_ooooee` (Appendix A).

Now suppose $OOOOEE$ is a cycle word at $n \geq 256$, and write $z = J^4(n)$ for the image after the prefix $OOOO$. Lemma 3.4(iv) gives $J(z) < (n+1)^2$, and the preceding even letter gives $z < (J(z) + 1)^2$, hence $z < (n+1)^4$. Lemma 3.3 on $OOOO$ gives $n^{81} \leq 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $OOOEEOE$ is a cycle word at $n \geq 256$. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor \sqrt{z_3} \rfloor$, so $z_3 < (y+1)^2$. Lemma 3.3 on OOO gives $n^{27} \leq 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 257$. We claim $(y+1)^3 < 2A^4$. If $y \leq A$, this is $(A+1)^3 < 2A^4$. If $y > A$, then $y \geq 258$, so $3y+1 < y^2$ and hence $(y+1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \leq yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. $\square$

Theorem 3.6 (small-cycle census). No word of length at most six is a cycle word at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word; every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start $n \geq 2$.

If every letter is odd, the start is odd, hence $n \geq 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \geq 3$, since $x^3 \geq (x+1)^2$. The orbit ascends strictly and never returns. Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word vE of the same length based at a cycle state $m \geq 2$. It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.2(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ($2 > 3^0$ and $4 > 3$).

Length three: the only expanding candidate is OOE . If $m \geq 5$ realizes OO , Lemma 3.4(i) gives $J^2(m) \geq (m+1)^2$, contradicting Lemma 3.4(iv). The only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd, so the final even letter is not realized.

Length four: the expanding filter requires three odd letters among the first three, leaving only O^3E , excluded by Lemma 3.4(v).

Length five: the filter requires four odd letters among the first four, leaving only O^4E , excluded by Lemma 3.4(v).

Length six: the filter requires at least four odd letters among the first five, leaving O^5E , $EOOOOE$, $OEEOOE$, $OOEOOE$, and $OOOOEE$. Lemma 3.4(v) excludes O^5E . The word $EOOOOE$ rotates one step onto $OOOOEE$, and $OEEOOE$ rotates two steps onto $OOEOOE$; both are excluded by Lemma 3.5.

It remains to exclude $OOEOOE$. Let this word be a cycle word at some start, and rotate to a cycle minimum $m \geq 2$. The three alignments of the two even letters are $OOEOOE$, $OEEOOE$, and $EOEOOE$. The last starts even, so it cannot be a minimum, by Theorem 3.2(ii). The middle starts OE : the first image is even and strictly below m^2 , whereas every even state after a cycle minimum is at least m^2 , as proved in Theorem 3.2(iii). Thus the minimum orientation is $OOEOOE$. In particular m is odd, so $m \geq 3$. The prefix OOE must be realized. If $m = 3$, then $3 \rightarrow 5 \rightarrow 11$ and 11 is odd, so OOE is not realized. Hence $m \geq 5$. Write $y = J^3(m)$ for the state after OOE . Then $y \geq m$ by minimality, so $y \geq 5$. The suffix OO is realized at y , and Lemma 3.4(i) gives $J^2(y) \geq (y+1)^2 \geq (m+1)^2$. The last letter is even, so Lemma 3.4(iv) gives $J^2(y) < (m+1)^2$. $\square$

Lemma 3.4(v) excludes every odd-run-then-even word O^aE with $a \geq 3$, of any length.

Lemma 3.7 (two length-seven exclusions). Neither $OOOOEOE$ nor $OOOOOEE$ is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 14$, then

$$n^{243} > 2^{422}(n+1)^{128}.$$

Indeed $14(n+1) \leq 15n$, so $(n+1)^{128} \leq (15/14)^{128}n^{128}$. The finite comparison $2^{422}15^{128} < 14^{243}$ then yields $2^{422}(n+1)^{128} < 14^{115}n^{128} \leq n^{243}$.

For $2 \leq n < 14$, neither word is realized: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooeoe` and `no_cycle_word_oooooee` (Appendix A).

Now suppose $OOOOOEE$ is a cycle word at $n \geq 14$, and write $z = J^5(n)$ for the image after the prefix $OOOOO$. Lemma 3.4(iv) on the last even letter, together with the preceding even letter, gives $z < (n+1)^4$. Lemma 3.3 on $OOOOO$ gives $n^{243} \leq 2^{422}z^{32}$, so $n^{243} < 2^{422}(n+1)^{128}$, contradicting the tail inequality.

Finally suppose $OOOOEOE$ is a cycle word at $n \geq 14$. Write $z_4 = J^4(n)$ and $y = J(z_4) = \lfloor \sqrt{z_4} \rfloor$, so $z_4 < (y+1)^2$. Lemma 3.3 on $OOOO$ gives $n^{81} \leq 2^{130}z_4^{16} < 2^{130}(y+1)^{32}$. Cubing yields $n^{243} < 2^{390}(y+1)^{96}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 15$. The same comparison $(y+1)^3 < 2A^4$ as in Lemma 3.5 holds at this smaller scale. Raising to the thirty-second power gives $(y+1)^{96} < 2^{32}(n+1)^{128}$. Combining with the cubed lower envelope produces again $n^{243} < 2^{422}(n+1)^{128}$. $\square$

Theorem 3.8 (small-cycle census through length seven). No word of length at most seven is a cycle word at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least eight.

Proof. The reduction of Theorem 3.6 applies at every length: an all-odd word cannot return, and every mixed cycle word rotates to an even-terminating cycle word based at a cycle state $m \geq 2$. Lengths at most six are Theorem 3.6. It remains to exclude even-terminating cycle words of length seven.

By Theorem 3.2(i) a cycle word is formally expanding. The even-terminating expanding length-seven words are exactly O^6E , EO^5E , OEO^4E , $OOEO^3E$, O^3EO^2E , O^4EOE , and O^5EE . Lemma 3.4(v) excludes O^6E . The word EO^5E rotates one step onto O^5EE , and OEO^4E starts OE , so it cannot be a cycle minimum (Theorem 3.2(iii)) and rotates two steps onto O^4EOE ; both leftovers are excluded by Lemma 3.7.

It remains to exclude $OOEO^3E$ and O^3EO^2E . Rotate either word to a cycle minimum $m \geq 2$. For $OOEO^3E$ the minimum orientation retains the internal even letter followed by the suffix OOO . Then $m \geq 3$, the prefix

through that even letter is realized, and Lemma 3.4(ii) at threshold 3 contradicts the last even cell. For O^3EO^2E the same bootstrap applies with suffix OO and threshold 5, once $m = 3$ is removed: at $m = 3$ the state after OOO is even, so the next even letter is not realized. $\square$

The census of Theorems 3.6 and 3.8 is a statement about period. The same cells organise leftover words by even-count. Write

$$e_a = 2(3^a - 2^a)$$

for $a \geq 0$.

Lemma 3.9 (trailing even run). If a cycle word based at n ends with $r \geq 1$ even letters, the state immediately before that run is strictly less than $(n+1)^{2^r}$.

Proof. The case $r = 1$ is Lemma 3.4(iv). Suppose the claim holds for some $r \geq 1$, and let vE^{r+1} be a cycle word at n . Write $z = J^{|v|}(n)$. The inductive hypothesis applied to the suffix E^r after the first of those even letters gives $J(z) < (n+1)^{2^r}$. The state z is even, so $z < (J(z) + 1)^2 \leq ((n+1)^{2^r})^2 = (n+1)^{2^{r+1}}$. $\square$

Lemma 3.10 (odd-run lower envelope). If $n \geq 1$ realizes O^a , then

$$n^{3^a} \leq 2^{e_a} J^a(n)^{2^a}.$$

Proof. Lemma 3.3 supplies a multiplicative constant C_v along any realized word, with $C_\varepsilon = 1$, $C \mapsto C \cdot 4^{2^j}$ on an even letter, and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j . On the pure odd word O^a every letter is odd, so $C_{O^{a+1}} = C_{O^a}^3 \cdot 4^{2^a}$. Writing $C_{O^a} = 2^{e_a}$ with $e_0 = 0$ yields the recurrence $e_{a+1} = 3e_a + 2^{a+1}$, because $4^{2^a} = 2^{2^{a+1}}$. The closed form $e_a = 2(3^a - 2^a)$ satisfies the recurrence and the initial value. $\square$

Lemma 3.11 (seven-odd window). No integer n with $2 \leq n < 256$ realizes the word O^7 .

Proof. This is a table of 254 seven-step evaluations: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_follows_seven_odds_of_lt256` (Appendix A). $\square$

Theorem 3.12 (two-even leftover families). Let $k \geq 6$ and $n \geq 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle word at n .

Proof. First, if $n \geq 256$, then

$$n^{3^{k-2}} > 2^{e_{k-2}}(n+1)^{2^k}.$$

The case $k = 6$ is the tail inequality of Lemma 3.5. If the display holds at some $k \geq 6$, cubing both sides produces

$$n^{3^{k-1}} > 2^{3e_{k-2}}(n+1)^{3 \cdot 2^k}.$$

The recurrence of Lemma 3.10 gives $e_{k-1} = 3e_{k-2} + 2^{k-1}$, so the desired comparison at length $k+1$ reduces to $2^{2^{k-1}} < (n+1)^{2^k}$. Equivalently $2 < (n+1)^2$, which holds for every $n \geq 256$.

Now suppose $O^{k-2}EE$ is a cycle word at such an n . Write $z = J^{k-2}(n)$. Lemma 3.9 with $r = 2$ gives $z < (n+1)^4$. Lemma 3.10 on the prefix O^{k-2} gives $n^{3^{k-2}} \leq 2^{e_{k-2}} z^{2^{k-2}}$, hence $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$, contradicting the tail.

Finally suppose $O^{k-3}EOE$ is a cycle word at such an n . Write $z = J^{k-3}(n)$ and $y = \lfloor \sqrt{z} \rfloor$, so $z < (y+1)^2$. Lemma 3.10 on O^{k-3} and cubing produce $n^{3^{k-2}} < 2^{3e_{k-3}}(y+1)^{3 \cdot 2^{k-2}}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. The comparison $(y+1)^3 < 2(n+1)^4$ of Lemma 3.5 applies at this scale. Raising it to the power 2^{k-2} and using $e_{k-2} = 3e_{k-3} + 2^{k-2}$ recovers again $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$.

For $2 \leq n < 256$, the cases $k = 6$ and $k = 7$ are Lemmas 3.5 and 3.7. The remaining short words O^6EE , O^5EOE , and O^6EOE fail to return on the same 254-start window; this is the Lean `native_decide` evaluation behind `no_cycle_word_two_even_ee` and `no_cycle_word_two_even_eoe` (Appendix A). Any longer leftover of either family begins with seven consecutive odd letters, which Lemma 3.11 forbids on this window. $\square$

A cycle word w at n is *minimum-based* when n is a cycle minimum: $J^j(n) \geq n$ for every $0 \leq j < |w|$. The remainder after a proper prefix of a cycle word need not itself be a cycle word at the prefix endpoint. The next statement therefore transports the tail inequality of Theorem 3.12, not the cycle-word exclusion at a later start.

Theorem 3.13 (first-even transport). Let $n \geq 2$. No minimum-based cycle word at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Write $y = J^{a+1}(n)$ for the state after the first even letter. Minimum-basedness gives $y \geq n$. In the first family the remainder after that letter is O^bEE with $b+2 \geq 6$; in the second it is O^bEOE with $b+3 \geq 6$. The trailing-even and last-odd cells of those remainders are measured against the cycle start n . Combined with Lemma 3.10 at the remainder start y , the same algebra as in Theorem 3.12 produces

$$y^{3^{\ell-2}} < 2^{e_{\ell-2}}(n+1)^{2^\ell} \leq 2^{e_{\ell-2}}(y+1)^{2^\ell},$$

where ℓ is the remainder length. If $y \geq 256$, the first paragraph of Theorem 3.12 supplies the opposite inequality at y .

If $y < 256$, then $n \leq y < 256$. A gapped leftover of total length at least 17 has $a \geq 7$ or $b \geq 7$, so either the prefix or the remainder realizes seven consecutive odd letters, contradicting Lemma 3.11. The finitely many short-gap words with $2 \leq a \leq 6$ and $b \leq 6$ fail to be minimum-based cycle words on the window $2 \leq n < 256$; this is the Lean `native_decide` evaluation behind `no_cycleMin_gapped_three_even_ee` and `no_cycleMin_gapped_three_even_eoe` (Appendix A). $\square$

The hypothesis that the start is a cycle minimum is essential. If $y < n$, the leftover cell is measured against a larger start and need not contradict the tail at y . In particular, Theorem 3.13 does not assert that those words fail to be cycle words at a non-minimum start. That upgrade is Theorem 3.21.

Theorem 3.14 (three trailing evens). Let $a \geq 6$ and $n \geq 2$. The word O^aEEE is not a cycle word at n .

Proof. Write $z = J^a(n)$. Lemma 3.9 with $r = 3$ gives $z < (n+1)^8$. Lemma 3.10 then yields $n^{3^a} < 2^{e_a}(n+1)^{2^{a+3}}$ on any such cycle word. For $n \geq 128$ the opposite comparison

$$n^{3^a} > 2^{e_a}(n+1)^{2^{a+3}}$$

holds. The case $a = 6$ is $n^{729} > 2^{1330}(n+1)^{512}$. Indeed, for $n \geq 128$ one has $(n+1)^{512} < (129/128)^{512}n^{512} < e^4n^{512} < 64n^{512}$, so the claimed bound reduces to $n^{217} > 2^{1336}$. Since $n \geq 128 = 2^7$, the left side is at least 2^{1519} . Cubing the $a = 6$ comparison produces the general case: the recurrence of Lemma 3.10 reduces the inductive step to $(n+1)^4 > 2$, which holds for every $n \geq 128$.

For $2 \leq n < 128$, the case $a = 6$ is a table of evaluations of `OOOOOOEEE`: at every such start the word fails to return. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooooooee` (Appendix A). For $a \geq 7$ the prefix contains seven consecutive odd letters, which Lemma 3.11 forbids on this window. $\square$

Theorem 3.15 (mixed bunched family EOEE). Let $a \geq 5$ and $n \geq 2$. The word O^aEOEE is not a cycle word at n .

Proof. First let $n \geq 4$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and $p = J(y)$. Lemma 3.9 with $r = 2$ after the prefix O^aEO gives $p < (n+1)^4$. The letter after y is odd, so Lemma 3.10 at length one yields $y^3 \leq 4p^2 < 4(n+1)^8$. For $n \geq 4$ one has $4(n+1)^8 < (n+1)^9$, hence $y < (n+1)^3$. The even cell at z then gives $z < (y+1)^2 \leq (n+1)^6$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{6 \cdot 2^a}.$$

For $a = 5$ and $n \geq 314$, the opposite comparison $n^{243} > 2^{422}(n+1)^{192}$ holds. The base instance is the finite inequality $2^{422}315^{192} < 314^{243}$. If the display holds at some $n \geq 1$, the elementary comparison $n(n+2) < (n+1)^2$ upgrades it to the same display at $n+1$. Cubing then produces the comparison at $a+1$, once $(n+1)^6 > 4$. In particular the case $a = 6$ already holds for every $n \geq 16$.

For $2 \leq n < 314$ and $a = 5$, and for $2 \leq n < 16$ and $a = 6$, the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eooo` (Appendix A). For $a \geq 7$ and $n < 256$, Lemma 3.11 applies. For $a \geq 6$ and $n \geq 16$, the tail of the previous paragraph applies. $\square$

Theorem 3.16 (mixed bunched family $EOOEE$). Let $a \geq 4$ and $n \geq 2$. The word $O^a EOOEE$ is not a cycle word at n .

Proof. First let $n \geq 32$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and p for the image after the prefix $O^a EOO$. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The two letters after y are odd, so Lemma 3.10 at length two yields $y^9 \leq 2^{10} p^4 < 2^{10} (n+1)^{16}$. For $n \geq 32$ one has $2^{10} < (n+1)^2$, hence $y < (n+1)^2$. The even cell at z then gives $z < (y+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a} (n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a + 2$.

For $2 \leq n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoooo` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.17 (mixed bunched family $EOOOEE$). Let $a \geq 3$ and $n \geq 2$. The word $O^a EOOOEE$ is not a cycle word at n .

Proof. First let $a \geq 4$ and $n \geq 3$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and p for the image after the prefix $O^a EOOO$. Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The three letters after y are odd, so Lemma 3.10 at length three yields $y^{27} \leq 2^{38} p^8 < 2^{38} (n+1)^{32}$. For $n \geq 3$ one has $2^{38} < (n+1)^{22}$, hence $y < (n+1)^2$. The even cell at z then gives $z < (y+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a} (n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a + 2$, already used in Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The same two-even cell gives $z < (y+1)^2$, so Lemma 3.10 at the prefix O^3 yields $n^{27} < 2^{38} (y+1)^{16}$. The three-odd envelope on y still gives $y^{27} < 2^{38} (n+1)^{32}$.

If $y < 39$, then $n^{27} < 2^{38} \cdot 39^{16}$. The numerical comparison $2^{38} \cdot 39^{16} < 24^{27}$ contradicts $n \geq 24$.

If $y \geq 39$, the numerical comparison $40^{27} < 2 \cdot 39^{27}$ upgrades to $(y+1)^{27} < 2y^{27}$, hence $(y+1)^{27} < 2^{39} (n+1)^{32}$. Cubing the prefix bound three times produces $n^{729} < 2^{1026} (y+1)^{432}$. Raising the successor bound to the sixteenth power produces $(y+1)^{432} < 2^{624} (n+1)^{512}$. Combining these displays yields $n^{729} < 2^{1650} (n+1)^{512}$. The opposite comparison holds at $n = 197$ and persists to every larger start by the elementary comparison $n(n+2) < (n+1)^2$ already used in Theorem 3.15.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooooo` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.18 (mixed bunched family $EEOE$). Let $a \geq 5$ and $n \geq 2$. The word $O^a EEOE$ is not a cycle word at n .

Proof. First let $n \geq 4$, and write $z = J^a(n)$ and y for the last odd letter of $O^a EEOE$. The suffix EOE is a cycle suffix, so the last-odd cube of Theorem 3.15 gives $y^3 < (n+1)^4$. The two letters between z and y are even, so $z < (y+1)^4$. For $n \geq 4$ the successor comparison $(y+1)^3 < 2(n+1)^4$ upgrades this to $z < (n+1)^6$. Combined with Lemma 3.10, any such cycle word would satisfy the same display as Theorem 3.15:

$$n^{3^a} < 2^{e_a} (n+1)^{6 \cdot 2^a}.$$

The opposite comparison is therefore the tail of Theorem 3.15: it holds for $a = 5$ and $n \geq 314$, and already for $a = 6$ and $n \geq 16$.

For $2 \leq n < 314$ and $a = 5$, and for $2 \leq n < 16$ and $a = 6$, the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eeoe` (Appendix A). For $a \geq 7$ and $n < 256$, Lemma 3.11 applies. For $a \geq 6$ and $n \geq 16$, the tail of the previous paragraph applies. $\square$

Theorem 3.19 (mixed bunched family $EOEOE$). Let $a \geq 4$ and $n \geq 2$. The word O^aEOEOE is not a cycle word at n .

Proof. First let $n \geq 32$, and write $z = J^a(n)$, $w = \lfloor \sqrt{z} \rfloor$, and y for the last odd letter. The suffix EOE again gives $y^3 < (n+1)^4$. The one-odd envelope on w yields $w^3 \leq 4s^2$, where s is the image after O^aEO . The last-odd cell and $n \geq 32$ upgrade this to $w < (n+1)^2$, hence $z < (w+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the shared tail already used in Theorem 3.16.

For $2 \leq n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.20 (mixed bunched family $EOOEOE$). Let $a \geq 3$ and $n \geq 2$. The word $O^aEOOEOE$ is not a cycle word at n .

Proof. First let $a \geq 4$ and $n \geq 4$, and write $z = J^a(n)$, $u = \lfloor \sqrt{z} \rfloor$, and y for the last odd letter. The suffix EOE gives $y^3 < (n+1)^4$, hence $(y+1)^3 < 2(n+1)^4$. The two letters after u are odd, so Lemma 3.10 at length two yields $u^9 \leq 2^{10}s^4 < 2^{10}(y+1)^8$. Cubing that display against the last-odd successor bound produces $u^{27} < 2^{38}(n+1)^{32}$. The same comparison as in Theorem 3.17 then gives $u < (n+1)^2$, hence $z < (u+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy the shared two-even tail of Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The prefix O^3 against $z < (u+1)^2$ yields $n^{27} < 2^{38}(u+1)^{16}$, and the two-odd plus last-odd geometry of the previous paragraph yields $u^{27} < 2^{38}(n+1)^{32}$. These are the same two displays as in the $a = 3$ case of Theorem 3.17, with u in place of y , and the same small/large split applies.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.21 (gapped leftovers as cycle words). Let $n \geq 2$. No cycle word at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Every cycle word has a minimum-based rotation. It is therefore enough to check that every cyclic shift of either word is an already-excluded cycle-minimum orientation.

Write w for the gapped word. In the first family, $|w| = a + b + 3$. The rotation by $k = 0$ is the original word, excluded as a cycle minimum by Theorem 3.13. The rotation by $k = a + 1$ is the bootstrap word O^bEEEO^aE . That word has an internal even letter and last gap at least 2. If $a \geq 3$, the last-gap threshold of Lemma 3.4 at OOO and $N = 3$ excludes it. If $a = 2$, the same lemma at OO and $N = 5$ excludes every start $n \geq 5$; the remaining odd start $n = 3$ does not realize four consecutive odd letters, so it cannot follow O^b for $b \geq 4$. The rotation by $k = a + b + 2$ begins with the last even letter of w , which Theorem 3.2 forbids at a cycle minimum. Every other rotation ends with an odd letter, likewise forbidden at a cycle minimum.

In the second family, $|w| = a + b + 4$. The same four classes appear: the original word is Theorem 3.13; the rotation by $k = a + 1$ is O^bEOEO^aE , excluded by the same last-gap thresholds, with the remaining start $n = 3$ and $a = 2$ either failing to realize four odds or, when $b = 3$, reaching 6 after $OOOE$ and then meeting an odd letter; the rotation by $k = a + b + 2$ begins OE , forbidden by Theorem 3.2; and every other rotation ends odd.

The original start need not be a cycle minimum. After rotation the start is a minimum, so the hypothesis $y < n$ that blocked Theorem 3.13 does not arise. $\square$

The word census of Theorems 3.6 and 3.8 stops at length seven; Section 5 excludes later periods by financing, not by a length-eight or length-nine word census. Theorem 3.12 excludes the two leftover orientations of length eight, but the census assembly of Theorem 3.8 is not extended. Length nine is the first even-terminating expanding length that admits three even letters; Theorems 3.13–3.21 treat infinite families of those leftovers, not every length-nine word. A single coarse successor power $(n+1)^K$ cannot exclude the

four families of Theorems 3.17–3.20 by the same cell used for Theorems 3.15 and 3.16: at the first expanding prefix length, the exponent $K \cdot 2^a$ meets or exceeds 3^a for those four words, so those arguments use a tight last-odd cell.

5. Cycle finance

A cycle word is formally expanding (Theorem 3.2), yet the orbit returns exactly. The multiplicative surplus $3^o - 2^L$ must be financed by the floor remainders, which are relatively $O(1/x)$ in logarithms. The resulting bound on the cycle minimum excludes every period that is not a near-convergent of $\ln 2 / \ln 3$, once a residual floor is known.

Throughout this section write $L = |w|$ and $o = \#O(w)$ for a cycle word w based at a cycle minimum $n \geq 2$. Natural logarithms are written $\log$. The unique one-step fibres of Section 3 give, for every state $x \geq 1$ with image $y = J(x)$,

$$y^2 \leq x^e < (y+1)^2, \quad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}. \end{cases}$$

Lemma (dyadic-cell logarithm). If $z, y \geq 1$ and $z < (y+1)^2$, then $\log z \leq 2 \log y + 2/y$.

Proof. The hypothesis gives $\log z \leq 2 \log(y+1)$. The inequality $\log(1+u) \leq u$ for $u > 0$ yields $\log(y+1) = \log y + \log(1+1/y) \leq \log y + 1/y$. $\square$

Lemma (one-step bounds). Let $x \geq 2$ and $y = J(x)$. If x is even, then $\log x \leq 2 \log y + 2/y$. If x is odd, then $3 \log x \leq 2 \log y + 2/y$.

Proof. The even cell is $y^2 \leq x < (y+1)^2$. The cell logarithm lemma on $z = x$ is the first claim. The odd cell is $y^2 \leq x^3 < (y+1)^2$. The same lemma on $z = x^3$ gives $\log(x^3) \leq 2 \log y + 2/y$. $\square$

On a cycle minimum every proper prefix is non-contracting: a prefix with $3^{\#O} < 2^k$ would satisfy $J^k(n) < n$ by Corollary 2.3, contradicting minimality. Thus $2^k \leq 3^{o_k}$ for every prefix of length k , where o_k is the odd count of that prefix.

Lemma (unrolled envelope). Write $x_k = J^k(n)$ and o_k for the odd count of the length- k prefix. For every $0 \leq k \leq L$,

$$3^{o_k} \log n \leq 2^k \log x_k + \frac{k 3^{o_k}}{n}.$$

Proof. The case $k = 0$ is an equality. Suppose the claim holds at $k < L$, and write $x = x_k$ and $y = x_{k+1}$. Minimality gives $n \leq y$, and the prefix law gives $2^{k+1} \leq 3^{o_{k+1}}$, hence

$$\frac{2^{k+1}}{y} \leq \frac{3^{o_{k+1}}}{n}.$$

If the next letter is even, then $o_{k+1} = o_k$ and the one-step bound gives $\log x \leq 2 \log y + 2/y$. Multiply by 2^k and add the inductive remainder:

$$3^{o_k} \log n \leq 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k 3^{o_k}}{n} \leq 2^{k+1} \log y + \frac{(k+1) 3^{o_k}}{n}.$$

If the next letter is odd, then $o_{k+1} = o_k + 1$. Multiply the inductive bound by 3 and apply the one-step bound in the form $3 \log x \leq 2 \log y + 2/y$:

$$3^{o_{k+1}} \log n \leq 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k 3^{o_k+1}}{n} \leq 2^{k+1} \log y + \frac{(k+1) 3^{o_{k+1}}}{n}.$$

This is the claim at $k+1$. $\square$

Theorem 5.1 (finance). Let w be a cycle word of length L with o odd letters, based at a cycle minimum $n \geq 2$. Then

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

Proof. Apply the unrolled envelope at $k = L$. Periodicity gives $x_L = n$ and $o_L = o$, so

$$3^o \log n \leq 2^L \log n + \frac{L \cdot 3^o}{n}.$$

The word is formally expanding, so $3^o > 2^L$. Rearranging and multiplying by n is the claim. $\square$

The only analytic input is $\log(1 + u) \leq u$. The Lean form is exactly Theorem 5.1 (constant 1).

The computational table of Theorem 5.3 uses a weaker per-step bound, valid on every cycle because every start below 12 reaches 1 (so every cycle state is at least 12). For a state x with image $y = J(x) \geq 12$, the relative defect $\delta = (x^e - y^2)/x^e$ satisfies $\delta \leq 2/y \leq 1/6$. Writing $\varepsilon = -\frac{1}{2} \log(1 - \delta)$ and using $-\log(1 - \delta) \leq \frac{6}{5} \delta$ on $[0, 1/6]$ gives $\varepsilon \leq (6/5)/y$. Unrolling $t_{i+1} = (e_i/2)t_i - \varepsilon_i$ around the cycle then yields

$$1 - \frac{2^L}{3^o} \leq \frac{6}{5} \sum_{i=1}^L \frac{1}{x_i \log x_i} \leq \frac{6}{5} \cdot \frac{L}{n \log n}.$$

This is Theorem 5.1 with an extra factor $6/5$ on the right, so it is conservative relative to constant 1.

The right-hand side is largest at the least admissible odd count $o_{\min}(L) = \min\{o : 3^o > 2^L\}$. Define

$$B(L) = \frac{6}{5} \cdot \frac{L \cdot 3^{o_{\min}(L)}}{3^{o_{\min}(L)} - 2^L}, \quad n_{\max}(L) = \max\{n \in \mathbb{N} : n \log n \leq B(L)\}.$$

Corollary 5.2. If every integer $2 \leq n \leq N_0$ reaches 1, then no nontrivial cycle of length L exists whenever $n_{\max}(L) \leq N_0$.

Proof. A periodic state never reaches 1, so every cycle state is at least $N_0 + 1$. Theorem 5.1 in the $6/5$ form forces the minimum to satisfy $n \log n \leq B(L)$, hence $n \leq n_{\max}(L)$. $\square$

Record values include $n_{\max}(19) = 297$, $n_{\max}(84) = 5599$, $n_{\max}(569) = 58398$, and $n_{\max}(1054) = 1997197$. The function $n_{\max}$ is large only when $3^{o_{\min}}/2^L$ is close to 1, i.e. when $o_{\min}/L$ is a one-sided approximation to $\log 2/\log 3$.

Theorem 5.3 (computational leftover). Every integer $2 \leq n \leq 10^6$ reaches 1. Consequently there is no nontrivial Juggler cycle of length at most 1053, and no cycle of any length $L \leq 10^5$ outside an explicit set of 397 near-convergent lengths of $\ln 2/\ln 3$. If a nontrivial cycle exists, its period is one of those 397 lengths, or at least 10^5 . The first record survivor is length 1054.

Proof. The floor is a first-passage descent induction: every start $2 \leq n \leq 10^6$ realizes a finite word with image strictly below the start, and strong induction on the image reaches 1. The longest first passage in the window has 253 steps (seed 78901); every iterate is an exact integer. Weisstein [5] records the same floor independently.

The gap table computes $o_{\min}(L)$ and $n_{\max}(L)$ by exact integer arithmetic for every $1 \leq L \leq 10^5$. Corollary 5.2 at $N_0 = 10^6$ excludes every L with $n_{\max}(L) \leq 10^6$. The surviving lengths in that range are exactly 397 near-convergents; the contiguous excluded prefix is $L \leq 1053$. The record (one-sided best-approximation) lengths in range are

$$1, 3, 11, 19, 84, 569, 1054, 25781, 50508,$$

with $n_{\max} = 3, 13, 52, 297, 5599, 58398, 1997197, 67410774, 420161535$. The first six of those already satisfy $n_{\max} \leq 10^6$ and are excluded. The first survivor is $L = 1054$. The remaining exceptions are multiples of 1054 and combinations of the convergent lengths; the full list is Appendix B. $\square$

The former record lengths 84 and 569 are therefore not live candidates at this floor. An independent Lean development, using the residual floor 261 and a census through length 19, concludes that the period is 84 or at least 85. That statement is recorded in Appendix A. Relative to Theorem 5.3 it is formalization lag, not a live leftover.

6. Remarks

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$.

Theorem 6.1 (uniform short certificates). Let $n \geq 2$.

1. If n is even, then the one-letter word E is a descent certificate: $J(n) = \lfloor \sqrt{n} \rfloor < n$.
2. If n is odd and $J(n)$ is even, then the two-letter word OE is a descent certificate.

Proof. For (1), $\lfloor \sqrt{n} \rfloor \leq \sqrt{n} < n$ for $n \geq 2$. For (2), the word OE is realized by hypothesis, and

$$J^2(n) = \lfloor \sqrt{\lfloor n^{3/2} \rfloor} \rfloor \leq n^{3/4} < n$$

for $n \geq 2$. $\square$

The starts not covered by Theorem 6.1 are exactly the odd-to-odd starts. In particular, if $n \geq 2$ has no descent certificate of any length, then n is odd-to-odd. The converse is false: many odd-to-odd starts descend after a longer word. If every start above 1 has some descent certificate, ordinary strong induction yields arrival at 1. That hypothesis is not proved.

The first odd-to-odd image expands, so ordinary strong induction cannot fire on the complement of Theorem 6.1. A uniform run bound on expanding blocks is likewise unavailable: four consecutive expanding blocks occur already at

$$1999 \xrightarrow{OOE} 5169 \xrightarrow{OOOOEE} 50093 \xrightarrow{OOE} 193753 \xrightarrow{OOE} 887471,$$

and the relative slack of a single letter tends to 0 with the state (Section 2).

No theorem forces every exact integer state into a contracting prefix. In particular, it is open whether every start reaches 1, and open whether a cycle of leftover length exists.

Lean proofs of the theorems of Sections 2, 3, 5, and 6 are in the project repository. From a clone, `lake build Problems.JugglerPaper` builds only the modules named in Appendix A. Theorem 5.3 is a named computation, not a Lean theorem.

Appendix A. Lean names

The proofs in the text are ordinary integer arguments. The following names are the corresponding Lean theorems in `formal/Problems/Juggler/`, imported by `Problems.JugglerPaper`.

Text	Lean
itinerary semantics	<code>follows_iff_word</code> , <code>image_eq_iterate</code> , <code>image_append</code>
Theorem 2.1	<code>image_monotone_of_follows</code>
Theorem 2.2	<code>power_bound_word</code>
Corollary 2.3	<code>power_bound_contracts</code>
Theorem 2.4	<code>global_defect_identity</code>
Theorem 2.5	<code>global_defect_eq_zero_iff_localsTight</code> , <code>global_defect_eq_zero_implies_monochrome</code> , <code>power_bound_eq_iff_extremal</code>
Theorem 2.6	<code>global_defect_append</code>
Corollary 2.7	<code>image_eq_start_defectRatio</code>
per-step slack	<code>one_plus_eta_lt_succ_sq</code>
Lemma 3.1	<code>odd_cell_unique</code>

Text	Lean
Theorem 3.2	cycle_word_formally_expanding, cycleMin_start_odd, cycleMax_start_even, cycleMin_not_end_odd, square_scale_superquadratic, cycleMin_to_even_superquadratic
Lemma 3.3	lower_growth_word
Lemma 3.4	oo_suffix_threshold, ooo_suffix_threshold, threshold_inherits_odd_append, cycle_last_even_interval, no_cycle_odd_run_append_even
Lemma 3.5	no_cycle_word_oooeoe, no_cycle_word_ooooee
Theorem 3.6	no_cycle_word_length_le_six
Lemma 3.7	no_cycle_word_ooooeoe, no_cycle_word_ooooeee
Theorem 3.8	no_cycle_word_length_le_seven, with no_cycle_word_ooeoooe, no_cycle_word_oooeooo
Lemma 3.9	cycle_trailing_evens_lt
Lemma 3.10	lowerDenom_replicate_odd, odd_run_lower_growth
Lemma 3.11	no_follows_seven_odds_of_lt256
Theorem 3.12	no_cycle_word_two_even_ee, no_cycle_word_two_even_eoe
Theorem 3.13	no_cycleMin_gapped_three_even_ee, no_cycleMin_gapped_three_even_eoe
Theorem 3.14	no_cycle_word_three_even_eee, with no_cycle_word_ooooooeee
Theorem 3.15	no_cycle_word_three_even_eooo
Theorem 3.16	no_cycle_word_three_even_eooee
Theorem 3.17	no_cycle_word_three_even_eooooe
Theorem 3.18	no_cycle_word_three_even_eoeoe
Theorem 3.19	no_cycle_word_three_even_eoeoe
Theorem 3.20	no_cycle_word_three_even_eooeoe
Theorem 3.21	no_cycle_word_gapped_three_even_ee, no_cycle_word_gapped_three_even_eoe
cell logarithm	log_le_two_log_add
one-step bounds	log_step_even, log_step_odd
unrolled envelope	cycleMin_log_envelope
Theorem 5.1	cycleMin_finance
Theorem 6.1	even_finiteProgress, odd_even_finiteProgress
no certificate $\Rightarrow$ odd-to-odd	no_finiteProgress_implies_odd_odd
induction to 1	reachesOne_of_all_finiteProgress
four-block chain	four_block_pe_1999

The following names are a Lean companion, not theorems of the text. They use a residual floor of 261 and a census through length 19. Relative to Theorem 5.3 they are formalization lag.

Companion	Lean
residual floor 261	reachesOne_of_lt_two_hundred_sixty_one
census through length 19	no_cycle_word_length_le_nineteen
period 84 or ≥ 85	cycle_word_length_eighty_four_or_ge_eighty_five
Eliahou packaging	cycle_word_eliahou_leftover

Appendix B. Exceptional lengths

The record near-convergents of $\ln 2 / \ln 3$ through length 10^5 , with the $6/5$ bound $n_{\max}$ of Section 5, are

L	$o_{\min}$	$n_{\max}$
1	1	3
3	2	13
11	7	52
19	12	297
84	53	5599
569	359	58398
1054	665	1997197
25781	16266	67410774
50508	31867	420161535

At the floor $N_0 = 10^6$ the first six rows are excluded. The surviving exceptional lengths through 10^5 number 397: the 94 multiples of 1054 in that range, together with combinations of the convergent lengths (for instance $23757 = 22 \cdot 1054 + 569$). The complete list is the `lengths` array of the object with `"floor": 1000000` in `data/research/juggler/cycle_finance/exceptions.json` in the repository of Appendix A. The gap table and the first-passage floor live in the same directory; they are regenerated by `python -m research.juggler_sequence.cycle_finance`.

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